Shraddha Harihar

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Education

- **M.Sc. Mathematics** — Aug 2023 - May 2025
  Dr. Vishwanath Karad MIT World Peace University, Pune, India — *GPA: 9.15/10*
- **B.Sc. (Hons) Mathematics** — Aug 2020 - May 2023
  M. S. Ramaiah University of Applied Sciences, Bangalore, India — *GPA: 8.78/10*

Research Experience

- **k-Domination Vertex Connectivity of Connected Graphs** — Aug 2023 - Dec 2024
  MIT World Peace University — *Pune, India*
  - **Concept Introduction**: Introduced and defined the novel parameter k-domination vertex connectivity, integrating graph domination and vertex connectivity.
  - **Mathematical Results**: Established theorems, exact formulas, and bounds for paths, cycles, and regular graphs using rigorous proof techniques.
  - **Computational Verification**: Developed Python code to verify results and explore concrete examples.
  - **Paper Submission**: Manuscript submitted to peer-reviewed journal for publication, currently under review.
- **Applications of Number Theory in Cryptography** — May 2023
  M. S. Ramaiah University of Applied Sciences — *Bangalore, India*
  - **Cryptographic Foundations**: Analyzed the foundational role of number theory (prime factorization, modular arithmetic, Discrete Logarithm Problem) in modern public-key cryptosystems like RSA and Diffie-Hellman.
  - **Cryptographic Security**: Investigated how security relies on the computational hardness of problems like integer factorization and the DLP.
  - **Algorithmic Foundations**: Explored applications of Euler's theorem, Miller–Rabin primality testing, and properties of primitive roots in cryptographic protocols and security proofs.
  - **MATLAB Implementation**: Implemented RSA encryption in MATLAB to demonstrate practical use of theoretical principles.

Conferences

- **Oral Presentation: RAPIAM-2025** — 5-6 June, 2025
  JSPM University in collaboration with the Indian Mathematical Society. — *Pune, India*
  - **Paper Title**: On the k-Domination Vertex Connectivity of Connected Graphs.

Submitted Manuscripts

- *On the k-Domination Vertex Connectivity of Connected Graphs.* Shraddha M. Harihar, Amol P. Narke, Prashant P. Malavadkar.

Skills

- **Mathematics**: Graph Theory, Combinatorics, Linear Algebra, Number Theory, Probability and Statistics, Cryptography.
- **Machine Learning**: Supervised and Unsupervised learning, Neural Networks, Mathematical ML Theory.
- **Programming**: Python, R, MATLAB, SQL.
- **Libraries and Frameworks**: NumPy, SciPy, Pandas, Matplotlib, TensorFlow, Keras, Scikit-learn, Seaborn, PyTorch.
- **Tools**: LaTeX, Jupyter Notebook, RStudio, SageMath, VS Code, Spyder, Git/Github.

Projects

- **Machine Learning Algorithms from Scratch (2024)**: Built core supervised learning algorithms from scratch in Python, covering Linear Regression, Logistic Regression, Decision Trees, and Kernel SVM. Focused on understanding the mathematical foundations of each method rather than just applying library implementations.
- **Tic-Tac-Toe vs AI: Minimax Algorithm (June 2024)**: Implemented the Minimax algorithm to build an unbeatable AI opponent, working through the full game tree search and optimal move selection at every state.

Work Experience

- **Research Analyst** Feb 2025 - Feb 2026
  MyAnatomy Integration
  - **AI/ML Research**: Investigated how AI and ML tools, including LLMs and NLP APIs, could be integrated into a recruitment platform, evaluating feasibility and fit.
  - **NLP Prototyping**: Built proof-of-concept systems for job-role extraction and automated question generation to test model performance in practice.
- **Data Science Intern** April 2024
  Prodigy Infotech
  - **Data Analysis**: Worked with customer data applying exploratory data analysis techniques and built a decision tree classifier.
  - **Sentiment Analysis**: Used NLP tools to analyze sentiment in customer reviews, turning unstructured text into measurable insights.

Achievements

- **Best Research Project Award**: Recognized for outstanding research contributions in the M.Sc. Mathematics program for work on k-domination vertex connectivity of connected graphs (MIT-WPU, May 2025).
- **Winner**: Ideathon, Hackathon (MIT-WPU, March 2024). Focused on object detection using YOLO algorithm.
- **Academic Excellence Awards, MIT-WPU**:
  - **The Dr. Vishwanath Karad scholarship (100% Merit):** Awarded to the top 1% students for academic excellence, 2023-2025.
  - **Silver Medal:** Awarded for attaining a top-tier GPA in M.Sc. Mathematics cohort, May 2025.